ALP-ttbar Toy MC v1.9: Does the Layer Arrangement Matter at Fixed Material? — Preregistration and Results

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Toy study motivated by Alimena et al., arXiv:2610.38932, which extends Rygaard et al., JHEP 10 (2023) 138

What this document contains. Page 1: this overview. Pages 2–4: v1.9 post-registration results (doi:10.5281/zenodo.24052153). Pages 5–8: v1.9 preregistration, frozen on GitHub and Zenodo before any v1.9 simulation or fit (doi:10.5281/zenodo.24048262). Both reports are reproduced unchanged.
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1. Why v1.9

For a 17 MeV ALP the displaced e^+e^- pair looks almost like a photon conversion in the tracker material. In this toy, reducing the material (v1.6) and measuring z at every layer (v1.8) lowered the 17 MeV loss from 127× to 27×, and v1.8 showed that the remaining limit is set mainly by multiple scattering. Next-generation tracker concepts also place the first layer closer to the beam and use more, thinner layers. v1.9 asks whether this **arrangement** helps when the **total material is kept fixed**.

2. The preregistered plan

• **Four geometries** at 3.8 T, a quarter of the nominal material, z at every layer: G0 = v1.8 (pixel layers at 29, 68, 109, 160 mm); G1 = first layer at 20 mm; G2 = eight pixel layers of half thickness; G3 = both. Total material identical in all four.
• **Analytic estimate in advance:** at most about 17% better r–z resolution for G2, almost none for G1, partly offset by more material bands in the veto.
• **Safeguards:** sanity gate (G0 must reproduce v1.8; identical total material), three validations (U1–U3), seven predictions written before running.

3. Results

geometry (3D fit, z at every layer)	17 MeV	50 MeV	100 MeV	200 MeV
G0 = v1.8	26.7×	8.83×	3.28×	1.09×
G1 first layer closer	26.7×	8.81×	3.28×	1.09×
G2 twice as many thinner layers	26.9×	8.51×	2.99×	1.21×
G3 both	26.9×	8.53×	3.02×	1.21×

• **The arrangement hardly matters.** More, thinner layers raise the r–z significance by about 19% for vertices inside the pixel volume, but only a quarter of 17 MeV ALP pairs decay there.
• **More layers cost signal:** eight material bands instead of four remove 58% instead of 53% of 17 MeV pairs in the material veto; the r–z information compensates almost exactly.
• **The closer first layer does nothing here**: only 4% of 17 MeV pairs decay inside 29 mm.
• **All seven predictions confirmed**; sanity gate and U1–U3 passed; no deviation in the method.

4. What the series now says (toy level)

Across v1.6–v1.9, the **amount** of material is the handle on the lightest ALPs; z at every layer adds to it; the magnetic field and the arrangement of the pixel layers do not. Next steps (each preregistered): a dimuon control study, the material veto itself, a joint 3D vertex fit.

Records

record	DOI / repository
v1.9 post-registration (pages 2–4)	https://doi.org/10.5281/zenodo.24052153
v1.9 preregistration (pages 5–8)	https://doi.org/10.5281/zenodo.24048262
code	github.com/wspaik5/ALP_ttbar_Toy_MC_v1.9_Postregistration_Layer_Geometry